CSE212: SOFTWARE DEVELOPMENT METHODOLOGIES

YEDITEPE UNIVERSITY

SPRING 2026

ASSIGNMENT 3 – DUE DATE MARCH 3RD, 2026

Below, you can find the description of your third assignment. Additionally, you can find the expected output of this assignment in the *Application Walkthrough* section.

As part of your third assignment, you are required to improve your existing *Online Shop Inventory Management System* by introducing **inheritance**, using **ArrayList** instead of C-style arrays, supporting both **physical and digital items**, and allowing multiple customer orders.

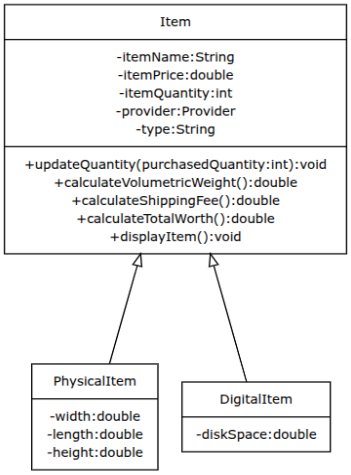

Fig. 1. *Item* class hierarchy.

First of all, you need to create a new class named *Provider* to store information about the supplier of an item. This class should include the following attributes:

- name (*String*)
- countryCode (*int*)
- areaCode (*int*)
- phoneNumber (*int*)

You may include helper (setter and getter) methods to retrieve and/or update provider information.

Then, you need to create a new class named *Item* that will act as a superclass for all inventory items - namely *PhysicalItem* and *DigitalItem* (see Fig. 1).

Your *Item* class, you should have the following attributes:

- itemName (*String*)
- itemPrice (*double*)
- itemQuantity (*int*)
- provider (*Provider*)
- type (*String*) → either "physical" or "digital"

You also need to implement the following methods in the *Item* class:

- void *updateQuantity (int purchasedQuantity)*: Updates the item quantity after an order is placed.

- double *calculateVolumetricWeight ()*:
  - If the item is **digital**, return 0.
  - If the item is **physical**, calculate:

  volumetricWeight = (length × width × height) / 5000 to compute the shipping fee later on

- double *calculateShippingFee()*:
  - If the item is **digital**, return 0.
  - If the item is **physical**, calculate shipping fee based on volumetric weight:
    - If volumetric weight > 3 → multiply by 0.5
    - Else if volumetric weight > 2 → multiply by 0.35
    - Else if volumetric weight > 1 → multiply by 0.25
    - Otherwise → shipping is free
  - Accordingly, the final formula to calculate the shipping fee is:

    shippingFee = itemPrice × volumetricWeight × rate

- double *calculateTotalWorth():* Reuse from your previous implementation.
- void *displayItem():* Reuse from your previous implementation.

Create a class named *PhysicalItem* that extends *Item*. In addition to the attributes and methods inherited from its superclass, *PhysicalItem* class should have three additional attributes:

- width (*double*)
- length (*double*)
- height (*double*)

These values will be used when calculating volumetric weight and shipping fee.

Also, create a class named *DigitalItem* that also extends *Item*. In addition to the attributes and methods inherited from its superclass, *DigitalItem* class should have one additional attribute:

- diskSpace (*double*): required storage space

For all *DigitalItem* instances, volumetricWeight and shippingFee are equal to 0.

The *Order* class should be initialized with two parameters - *Item* and its *type* (*String*). This class should have setter/getter methods to set and get the item ordered and its type. The type could either be digital or physical. Last but not least, this class should also have a *calculateOrderCost()* method that will calculate the total cost of an order, including item price and any applicable shipping fees.

As you have been introduced with ArrayLists, from now on you are required to store all items (both for physical and digital) and orders in a relevant type of an ArrayList. With this update, the inventory owner should now be able to add an endless number of items and customers must be able to place as many orders as they want. Accordingly:

- The system must support **adding multiple items**.
- The system must support **placing multiple orders**.
- Before placing an order, your implementation should:
    - check whether the item exists,
    - check whether sufficient quantity is available.

After a successful order:

- item quantity in the inventory must be updated accordingly.

PS: Do **NOT** store your item data in C-style arrays anymore, **DO** use ArrayLists. Test your program thoroughly to ensure it meets the requirements of the assignment.

INVENTORY MENU:

1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory

CUSTOMER MENU:

5. Add an order
6. Calculate order cost
7. Exit

Application Walkthrough

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INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 1

Item's Information:
Enter item name: iPhone
Enter item price: $999
Enter item quantity: 500

Provider's Information:
Enter provider name: Amazon
Enter country code: 90
Enter area code: 850
Enter phone number: 3902153

Physical or Digital Item? Physical
Enter width: 9
Enter length: 14
Enter height: 8
Physical item added successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
```

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6. Calculate Order Cost
7. Exit
Enter your choice: 1

Item's Information:
Enter item name: Baldur's Gate Game
Enter item price: $59,99
Enter item quantity: 30

Provider's Information:
Enter provider name: eBay
Enter country code: 01
Enter area code: 408
Enter phone number: 5168811

Physical or Digital Item? Digital
Enter disk space: 6
Digital item added successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 1

Item's Information:
Enter item name: Macbook
Enter item price: $1999
Enter item quantity: 230

Provider's Information:
Enter provider name: Apple
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Enter country code: 90
Enter area code: 800
Enter phone number: 4488298

Physical or Digital Item? Physical
Enter width: 13
Enter length: 18
Enter height: 3
Physical item added successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 2
Item : iPhone
Price: $999.0
Quantity: 500
Provider: Amazon
Provider Contact: +90(850)3902153

Item : Baldur's Gate Game
Price: $59.99
Quantity: 30
Provider: eBay
Provider Contact: +1(408)5168811

Item : Macbook
Price: $1999.0
Quantity: 230
Provider: Apple
Provider Contact: +90(800)4488298
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INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 3
Total number of items: 3

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 4
Total Worth of Item iPhone: $499500.0
Total Worth of Item Baldur's Gate Game: $1799.7
Total Worth of Item Macbook: $459770.0

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 5
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Enter the item name you want to order: Macbook
Enter the quantity: 30
Order has been placed successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 5
Enter the item name you want to order: Baldur's Gate Game
Enter the quantity: 5
Order has been placed successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 2
Item : iPhone
Price: $999.0
Quantity: 500
Provider: Amazon
Provider Contact: +90(850)3902153

Item : Baldur's Gate Game
Price: $59.99
Quantity: 25
```

```
Provider: eBay
Provider Contact: +1(408)5168811

Item : Macbook
Price: $1999.0
Quantity: 200
Provider: Apple
Provider Contact: +90(800)4488298

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 6
The total cost of Macbook order is: 59970.0
The total cost of Baldur's Gate Game order is: 299.95

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 7
Exiting... Goodbye!
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Submit your assignment folder in a zip file [class files will not be marked**], which has your student number as name, through the YULEARN (https://yulearn.yeditepe.edu.tr) latest by the end of Friday, April 3rd, 2026.**